



**An End to end platform for any waste management ecosystem to
fast forward waste through all touchpoints**

Teamstarbase Pvt Ltd





In the Cities



In the mountains



In the Oceans



In the rivers

Waste Everywhere

3196 landfills

But Why?

Current scenario

In India, About 43 million tonnes (70%) are collected, of which about 12 million tonnes are treated, and 31 million tonnes are dumped in landfill sites. With changing consumption patterns and rapid economic growth, it is estimated that urban municipal solid waste generation will increase to 165 million tonnes in 2030.



Waste infrastructures like collectors and recyclers lack real time data or the know how to optimize operations .



Waste infrastructures like MRF has no access to real time data on volume of waste handled impacting their operations.

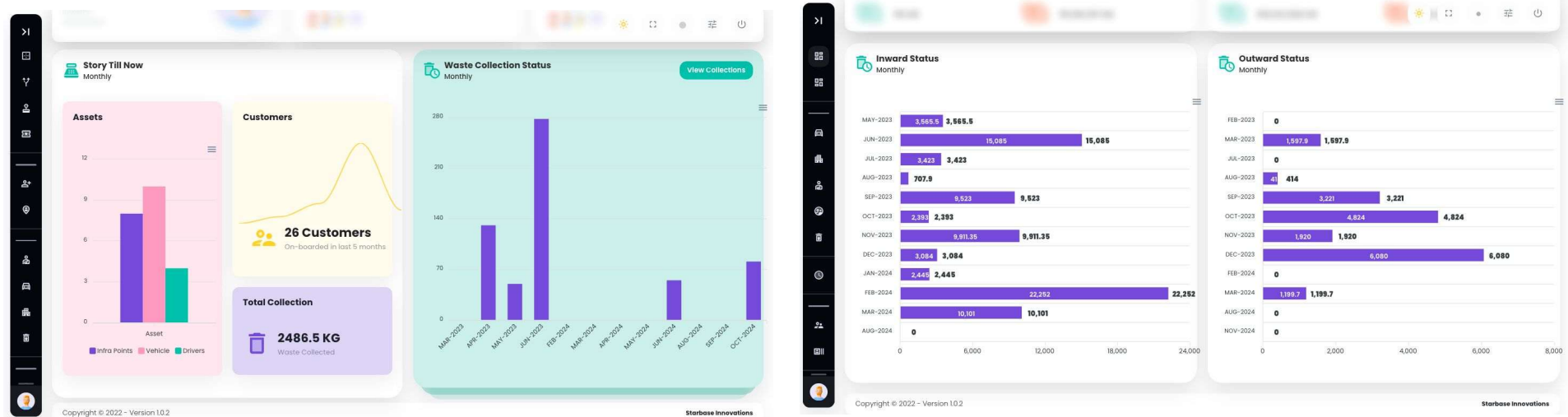


Govt. departments- face challenges to trace the movement of waste as there is no accountability and actionable data

The non availability of systems to capture real time accurate data in the waste management ecosystem leading to various forms of inefficiencies .

Solution

Swastha is an End-to-End platform to track, control and optimize the waste-collection, segregation and recycling process for all stakeholders in the ecosystem.



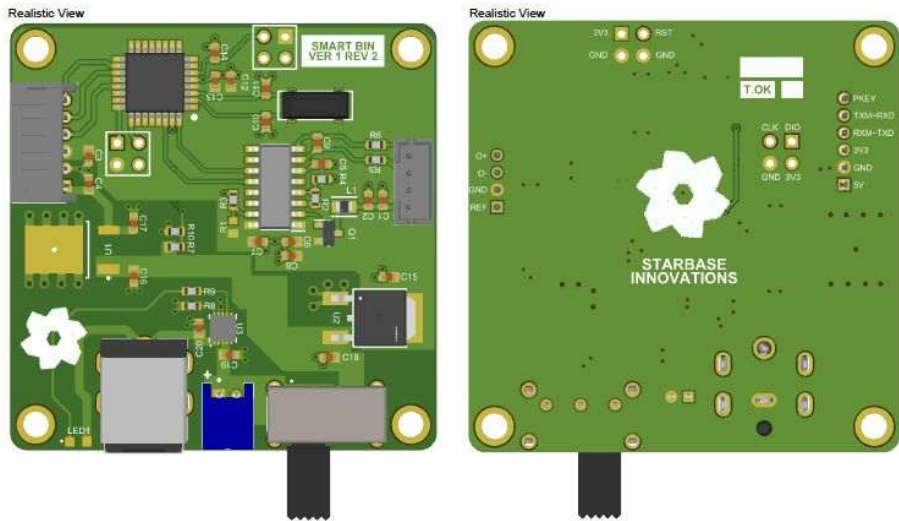
Collect Module : For medium and large scale waste collectors to track, control and analyze their operations and gain visibility to key stakeholders in the supply chain.

Recycle Module: For medium and large scale waste recyclers to track, control and analyze their operations and acquire EPR credit through PCB Portal.

Solution

Hardware

IOT Kit for weighing scale



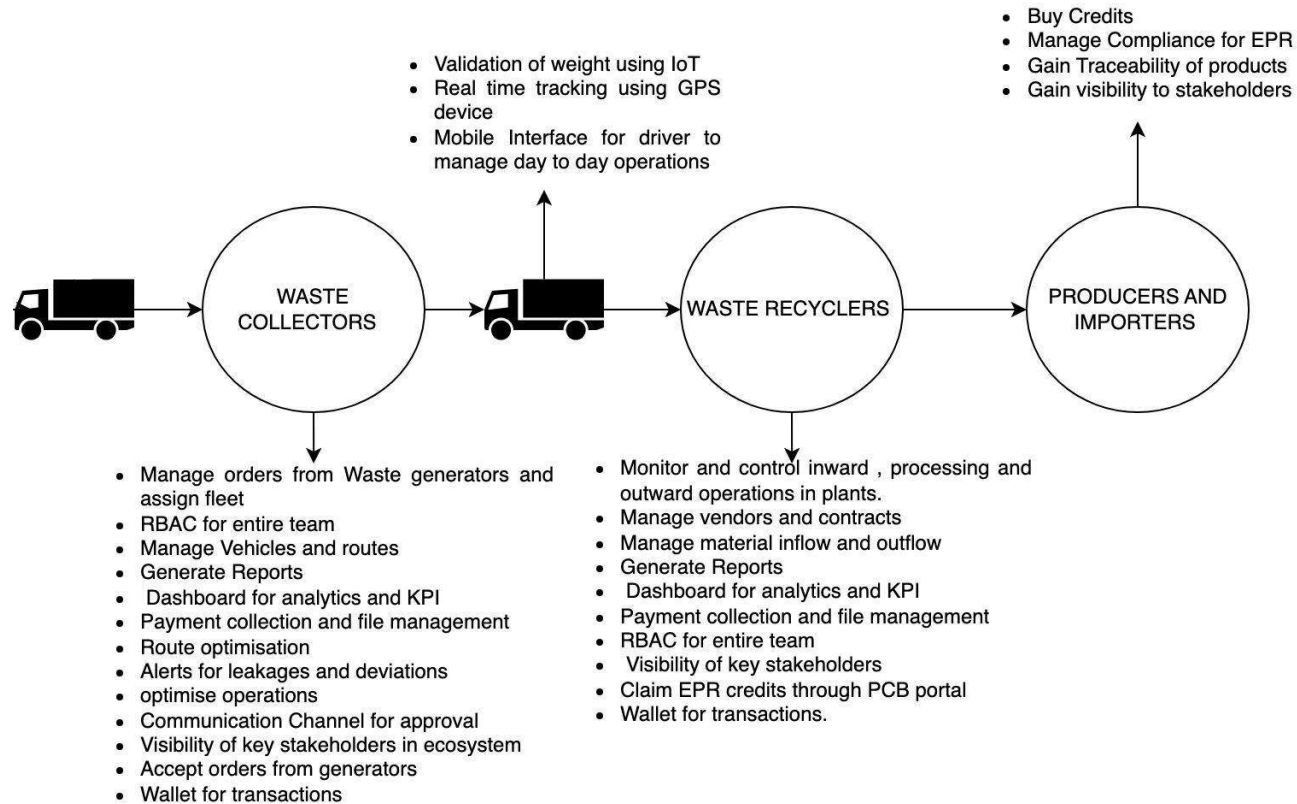
The IoT board developed is capable of connecting any industrial weighing system to the cloud to validate weight of waste during transaction between users. The board is designed and needs commercialisation

Weight Validation



The strain gauge sensor can be integrated to vehicles to track leakages. The MVP version of the sensor developed and validated.

How it works



Value Proposition

Waste Collectors : CREDAI clean city
And Green worms



- Track and control day to day operations
- Visibility of waste collectors and recyclers within ecosystem
- increase revenue by identifying new customers
- Increase efficiency by optimizing routes and operations
- Prevent leakage during transportation using IoT
- Manage payment collection

Waste Recyclers: Orion Recyclers



- Track and control day to day operations
- Visibility of raw material available in ecosystem
- Generate Reports on inward and outward operations to claim EPR credits
- Validate weight at every transaction using IoT
- Increase efficiency by maximizing capacity
- Increase Revenue through monetization of EPR Credit

State Government : Govt of Kerala



- Track movement of waste
- enforce accountability of waste generated and transported
- enable monitoring and action through existing login for all hierarchy

Market Opportunity

In Kerala Alone to address 90 Lakh waste generators

- 43 Dry waste Lifting agencies
- 1054 Dry waste transporters
- 1083 food waste collectors
- 250+ private agencies for other category
- 10+ medical waste collectors
- 178 registered plastic recyclers
- 489 Pig farms to handle food waste
- 1203 MCF
- 189 RRF
- 250 + recyclers
- 4000 + unorganised aggregators

Sources: MOHUA, Indian Pollution Board,
Haithamithram, LSGD Kerala MCA

Indian Market

TAM:
₹ 73,570 CRORES

**Starbase India
Opportunity:**
₹ 800 CRORES

Revenue Model

60% Gross Margins

Business Model

Pricing Per Month

Waste Collectors

Subscription Model

Per vehicle : INR 450
Per admin user : INR 2800
Per Operation User : INR 1200

Waste Segregation centers / MRFs

Subscription Model

Per Admin web user : INR 1800
Per plant Operator: INR 900

Waste Recyclers

Subscription Model and
Commission on EPR credits
generated

Per Admin web user : INR 1800
Per plant Operator: INR 900
Commission on EPR credits
generated : 15%

Weight Validation Kits

Sale of Hardware

Per IOT Kit : INR 2500
Per Vehicle weight validation :
INR 20000

Traction

Waste Handled 3 Kinds of waste	Govt Mandate 126 No of Vehicles on the platform	Govt Mandate ₹ 904,00,000 Won a Govt Order to develop a routing system
6 Number of Waste Collectors on the platform	1000 MT Tonnage moved	139 Resource Recovery Facilities Handled
	₹ 48,500 Current Monthly Recurring Revenue	1203 Govt mandate to connect infrastructure points to recyclers

GTM

Waste Collectors

Waste Segregation
centers / MRFs

Waste Recyclers

EPR

Go to market Strategy

Kerala is the focused geography .we plan to reach out to potential customers via industry associations , Govt partnerships , visiting them in person and via networking events. The contract signed by Govt of Kerala to implement Route tracking system will help us to penetrate the users involved in

- Lifting operation In Kerala
- Transportation operation in kerala
- Door to door collection agencies
- Recyclers

We plan to develop a open source platform to waste generators to discover waste collection businesses and contact them.

There are at present 178 registered plastic recyclers in Kerala PCB portal and 200+ unregistered users who would be targeted through field sales.

Competition

Competitor	Starbase	Satma CE	Recykal	Recircle
End to End Tracking	Available	Available	NA	NA
Weight validation	Available	NA	NA	NA
Stakeholder Visibility	Available	NA	Available	NA
EPR Credits	Available	NA	Available	Available

Team



Nithin Anil
Founder,

PG In Embedded and VLSI
Design
10 Years of Experience as
Serial Entrepreneur



Akhil Jayan
CTO

PG In System Design
10 Years of Experience in
Building and Implementing
Tech Products



Ahana K K
Co-Founder,
CSO

Master In Social Work
5 Years of Experience in
working on SDG Goals and
Business Development

ADVISORS:



Pankaj Thakar, Chief Mentor at
Padup Ventures

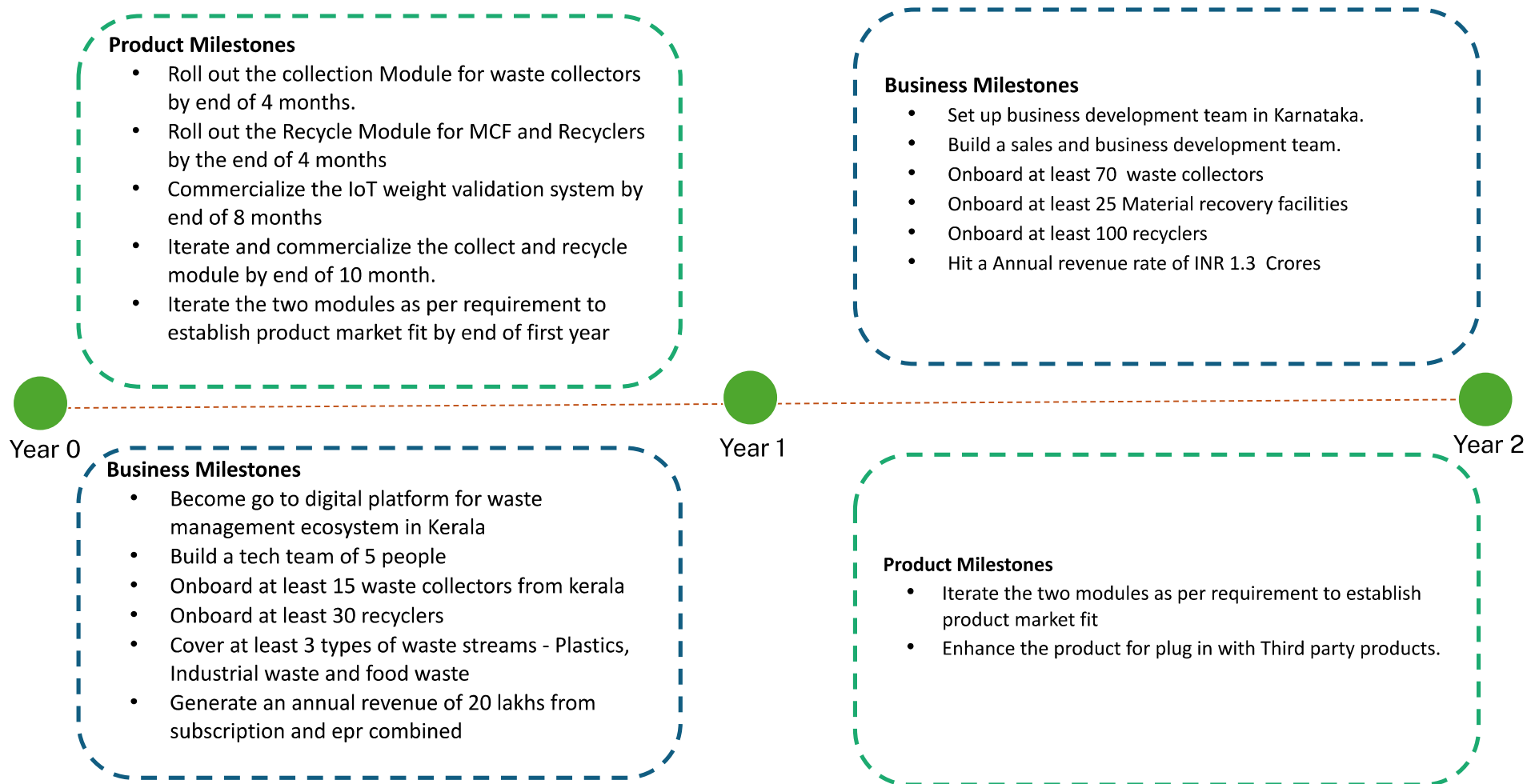


Dhananjay Gulwadi, Ex COO of
Assarian(SSW) group

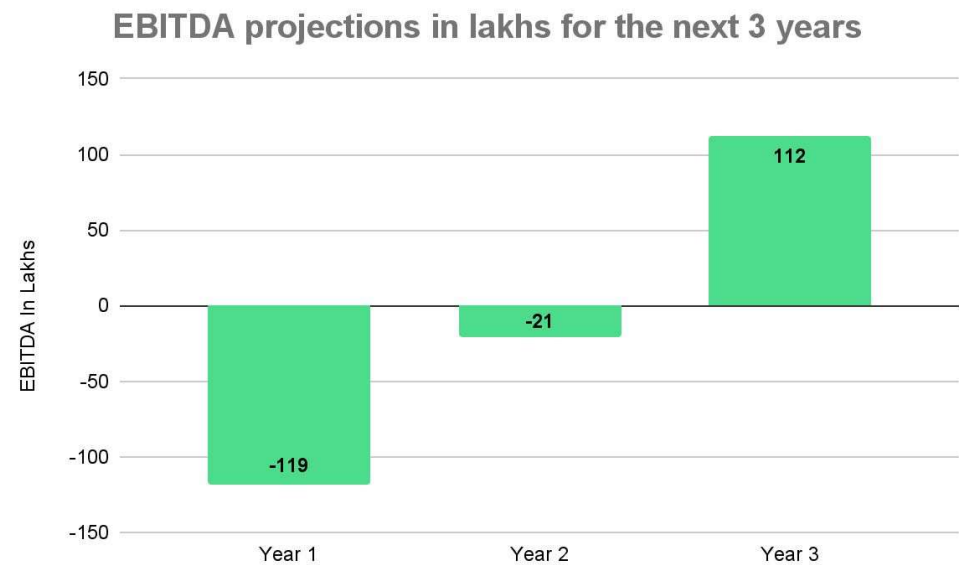


Teamstarbase Pvt Ltd

Growth Strategy For the Next 2 Years



Financial Projections



Ask & Utilization

Cap Table
95% with founders
5% PadUp
X% someone else

**15 Lakhs from AIC-IIITK
and 2 lakhs from CUSAT**
Past Fundraising Details/
Investors/ Valuations

Ask : 1.5 Crores @ 10% dilution

Technology Development [70 Lakhs]	Go to Market [40 Lakhs]
Working Capital [20 Lakhs]	Capital Expenditure [20 Lakhs]

Achievements

Climate Cohort UnLtd
India



CUSAT RUSA FUND
2024



UNICEF Innovation
Fund Finals



GEX Conclave Product
Showcase



ITU SEED
Acceleration
Program



AIC raise
incubation
program



USAID Finalist



Waste Collection
Digitization Contract
Government of Kerala



CREDAI Waste
Collection Digitization



WASH Innovation
Challenge



Greenr
Sustainability
Accelerator





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